GREEN TEAM DOCUMENTATION

Team #14: Kansas State HackCats

Table of Contents

Accessing Machines …….. 2

RDP ……. 2

SSH ……. 4

Web Interfaces…….. 6

Public Website…… 6

Admin Page……. 11

Ticketing System…… 14

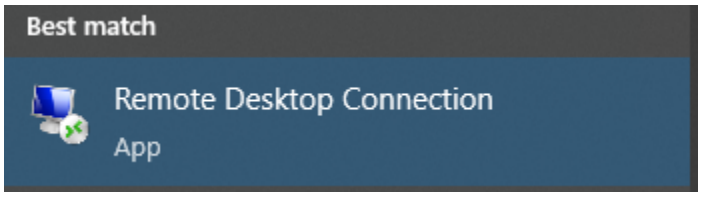
Accessing Machines

Logging into Windows machines with RDP

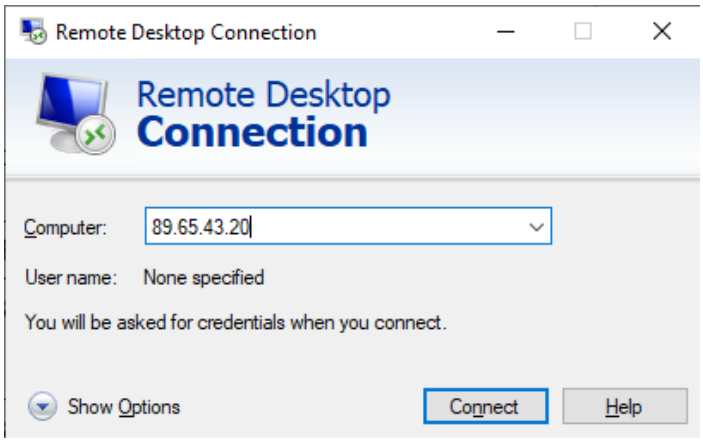
Administration of Windows Server Machines can be done using RDP (Remote Desktop Protocol). Below is a guide for remotely logging in using RDP (Note: Must be done on the competition network or VPN):

1. Click into the search bar at the bottom left of the screen to search for programs

2. Begin typing “Remote Desktop Connection” and click the Remote Desktop Connection program to launch an RDP window



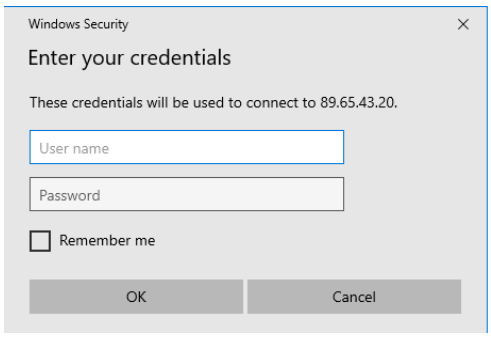
3. Enter the hostname or IP of the computer to access in the box provided.



Note: The following machines have RDP access allowed on the competition network:

Device	IP
Active Directory	89.65.43.10
Ticketing Machine	89.65.43.20
Admin Dashboard	89.65.43.40

4. When prompted, enter admin username & password for the target device



5. After clicking OK, RDP will launch a new window which will allow remote use and administration of the target Windows device

Logging into Linux machines with SSH

Administration of Linux server machines can be done using SSH (secure shell). This can be done in a terminal, or with a tool such as PuTTY or MobaXterm. To SSH into a machine, you must be connected to the VPN or competition network. Below is a guide on how to connect to a Linux server using all three:

The following hosts have SSH enabled:

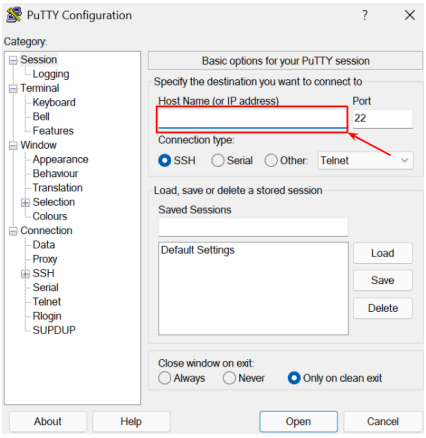
Device	IP
Management	89.65.43.30
Web	89.65.43.50

<u>Terminal:</u>

1. Enter this command, replacing “username” with the desired username and “ip-or-hostname” with the desired IP or host name:
   ```
   ssh username@ip-or-hostname
   ```
2. Enter password when prompted
3. If the password and username are correct, you will connect via SSH to the machine located at that IP address.

<u>PuTTY:</u>

1. Launch PuTTY, and enter the IP or the Host Name in the box labeled below:

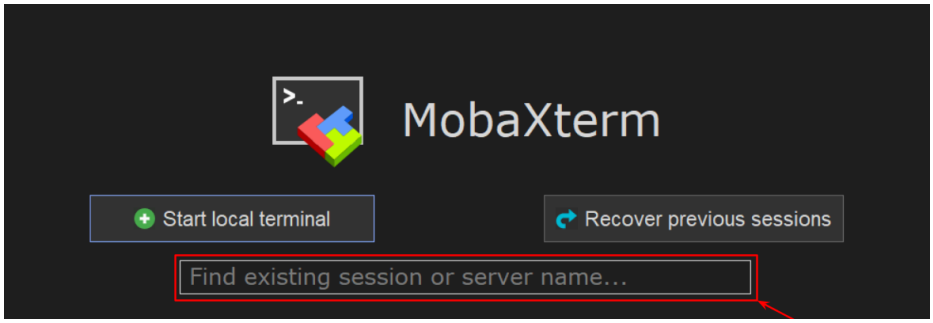


2. After entering the destination, click “Open”. You may be prompted to accept the host key to PuTTY’s cache. If you trust this machine, click “Accept”.

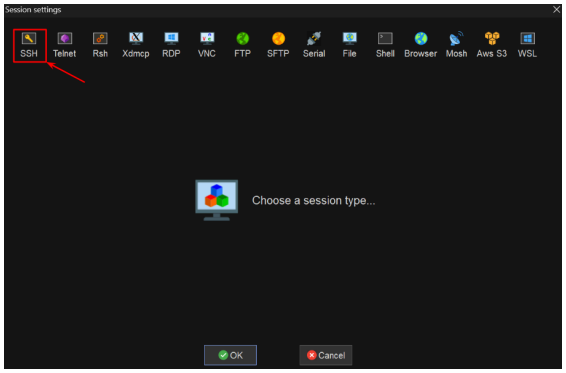
3. A terminal emulator will appear upon a successful connection, prompting you to enter the username, followed by the password. If these are correct, you will connect via SSH to the machine.

<u>MobaXterm:</u>

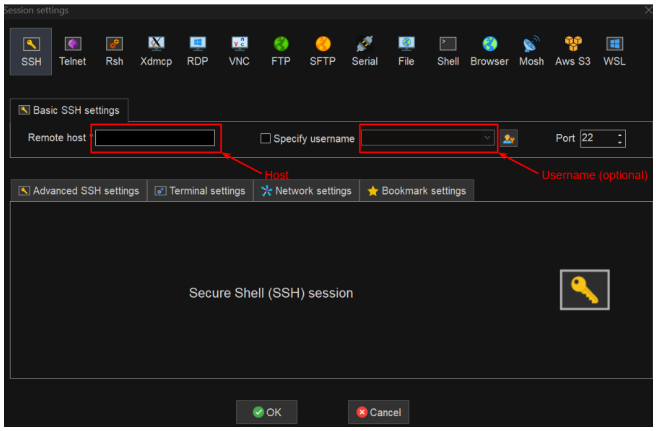
1. Launch MobaXterm, and enter the IP or Host Name in the box labeled below:



2. Select the connection type. In this case, it is SSH, labeled below.



3. The Remote host should be in the box labeled “Remote host”, and you can optionally add the username in the box labeled “Specify username”. Afterwards, click “OK”, and you will be prompted for a username if you did not specify one, and a password. If these are correct, you will be connected to the host via SSH.

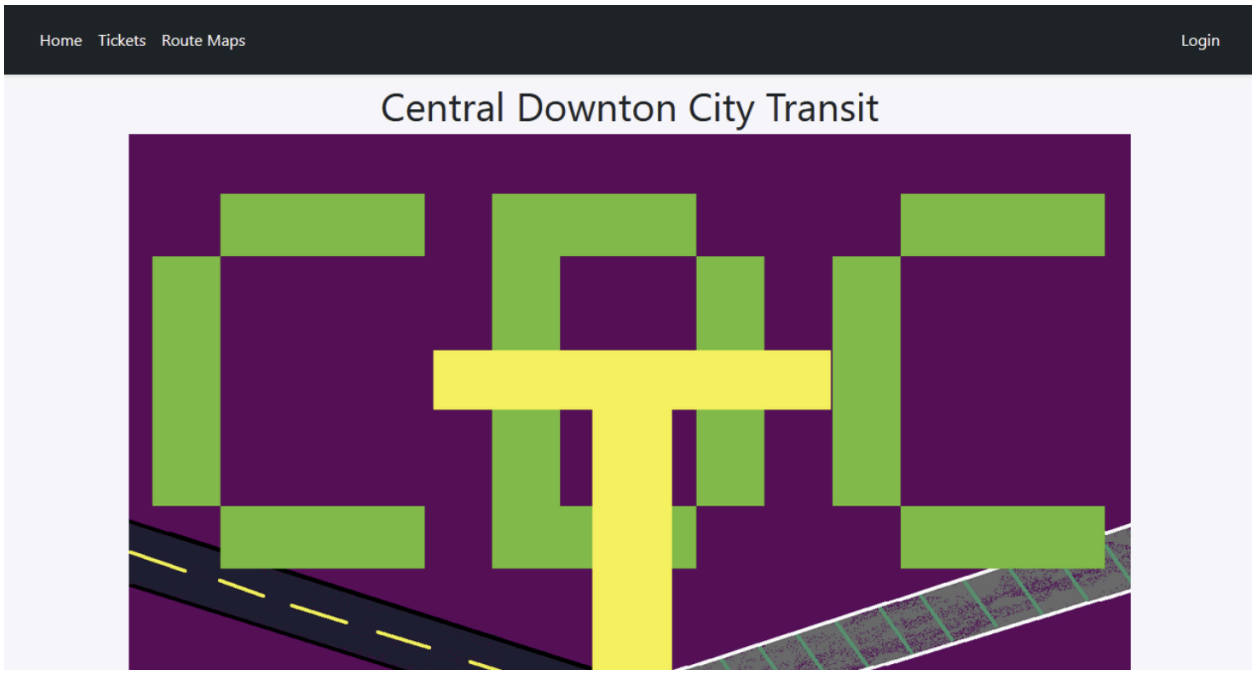


Web Interfaces

Public Web Site

The public website allows users to purchase tickets and track the status of the transit system.

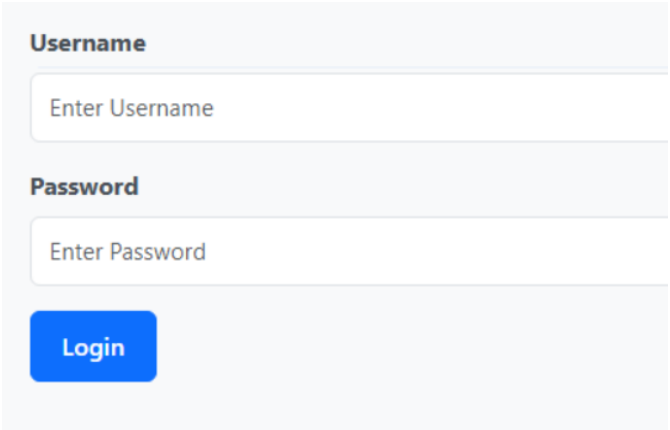
Proceed to www.team14.isucdc.com



Login:

1. Select the Login option on the top right of the screen.

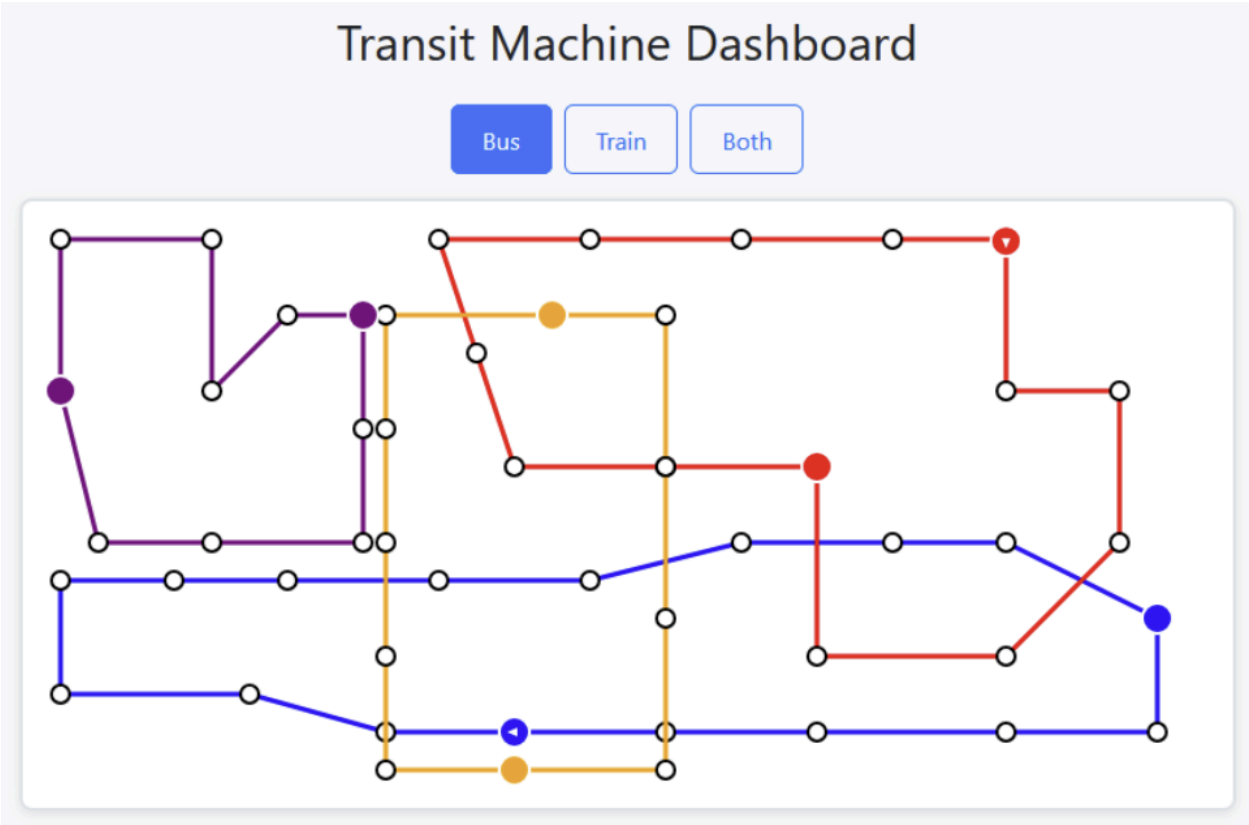
2. Enter username and password and click login



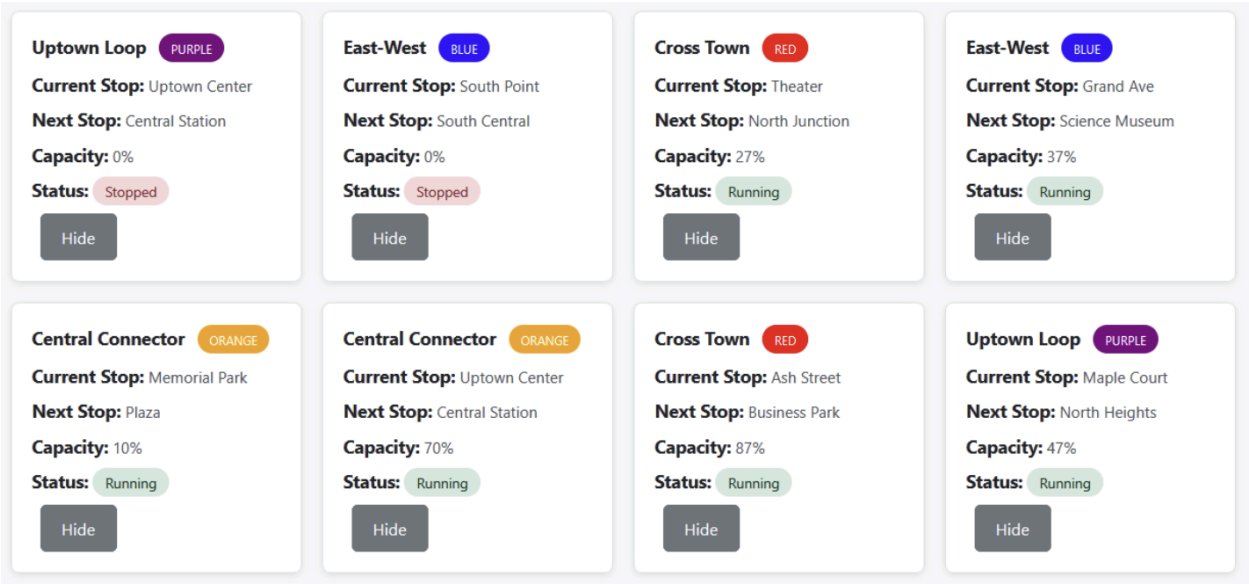
3. If login is successful, your username should now be displayed at the top right of the navigation bar

Route Maps

1. Select the “Route Maps” option in the top left of the header.
2. Select the desired option to see Bus, Train, or Both. This will pull up a map of routes as well as the current location of the bus or train.

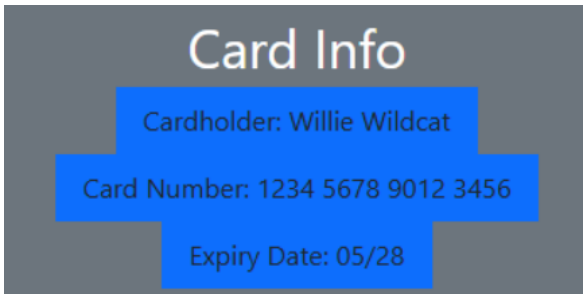


3. A summary of the buses or trains is further down. It details the current and next stop, available capacity, and running status of each transit option. Toggle the option's corresponding dot on the map on and off by selecting Show or Hide on the summary.

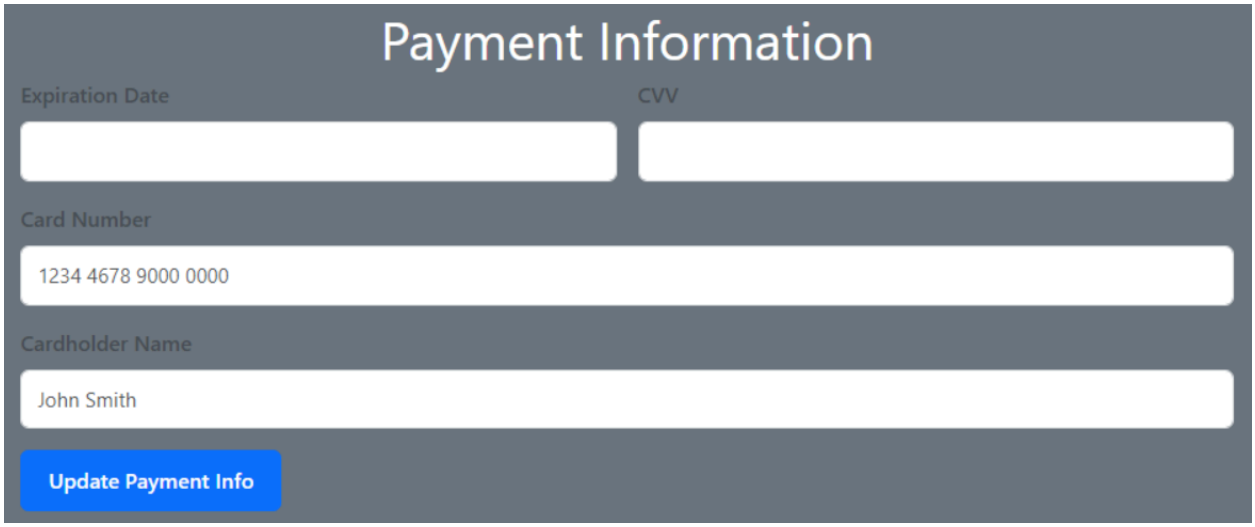


Modifying/Viewing Account Payment Information

1. Click on your username at the top right of the navigation bar
2. Current card information is displayed at the bottom of the page

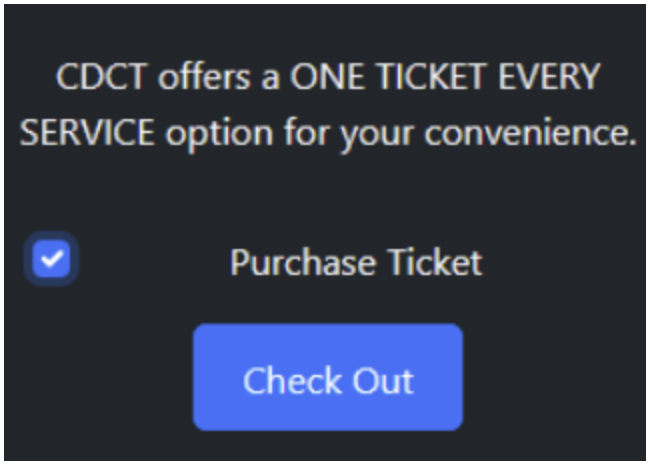


3. To update your payment information, enter the new card information in the input boxes and click ‘update payment information’



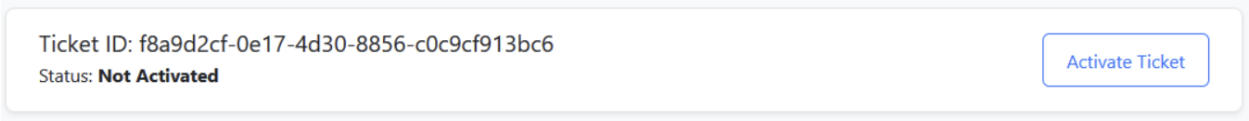
Purchase Tickets:

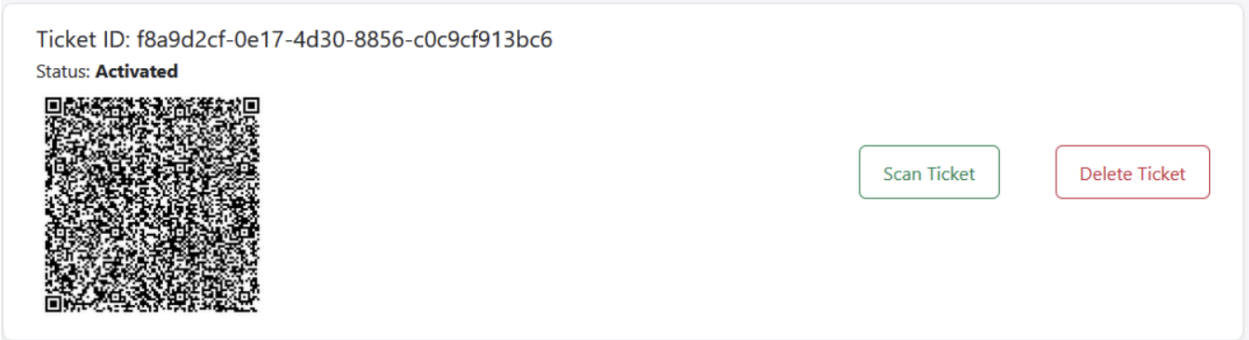
1. Select the Tickets option on the top left of the header. Check the box next to “Purchase Ticket” and click the “Check Out” button.



2. This should generate the below popup. If an error occurs displaying ‘no payment method’, check your account to verify you have entered a valid payment method.

3. The ticket will now be available on your tickets dashboard. Click “Activate Ticket” to activate your ticket and display the QR code to be scanned.

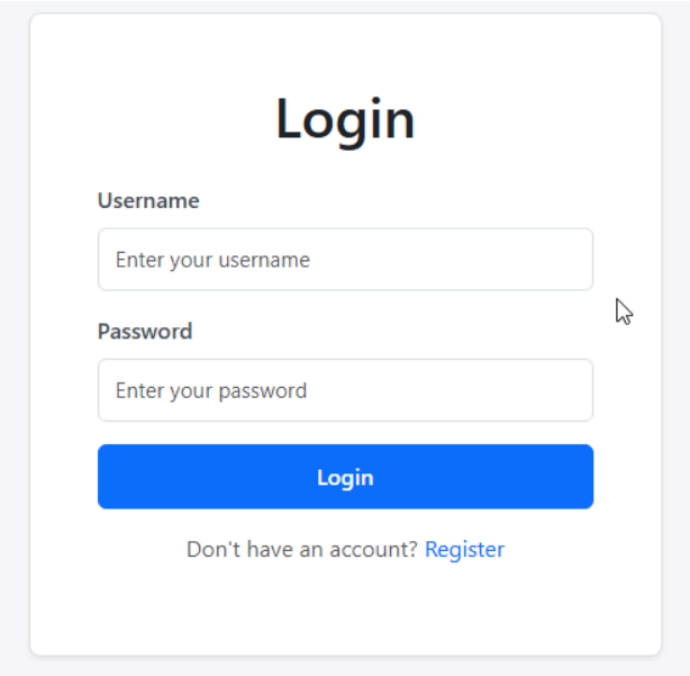




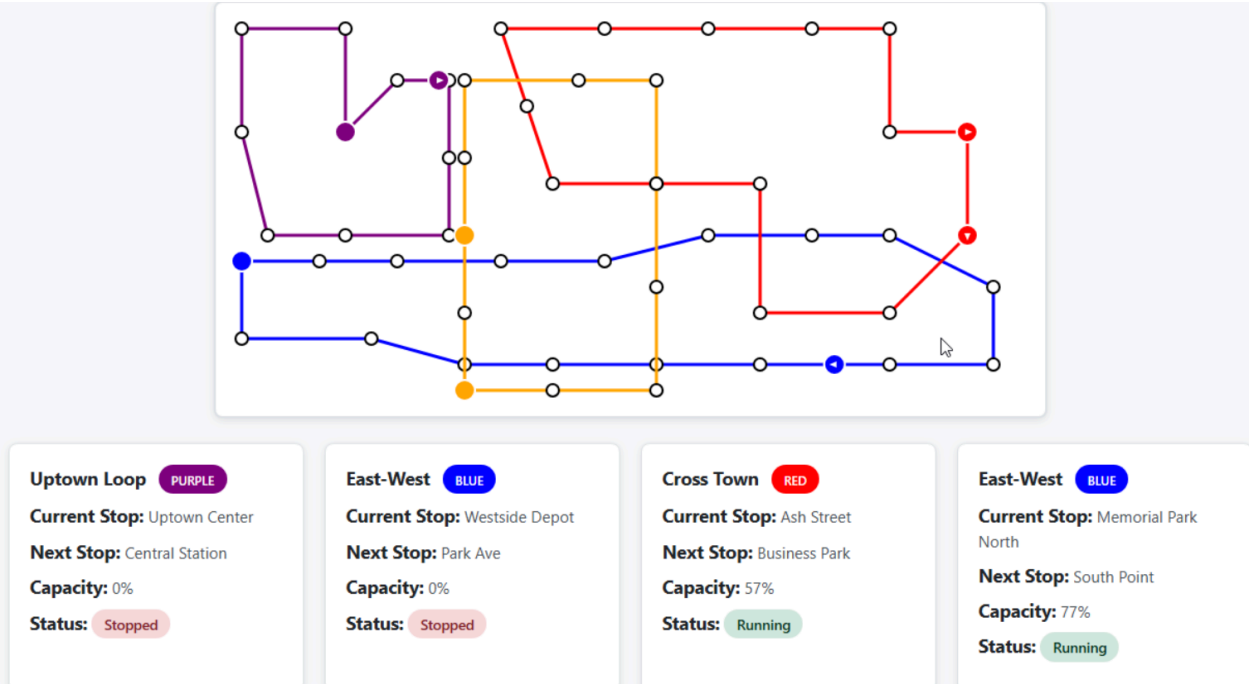
Admin Page

The admin dashboard allows for transit administrators to view and modify data related to the bus and train system.

1. Navigate to admin.team14.isucdc.com and enter your administrator credentials

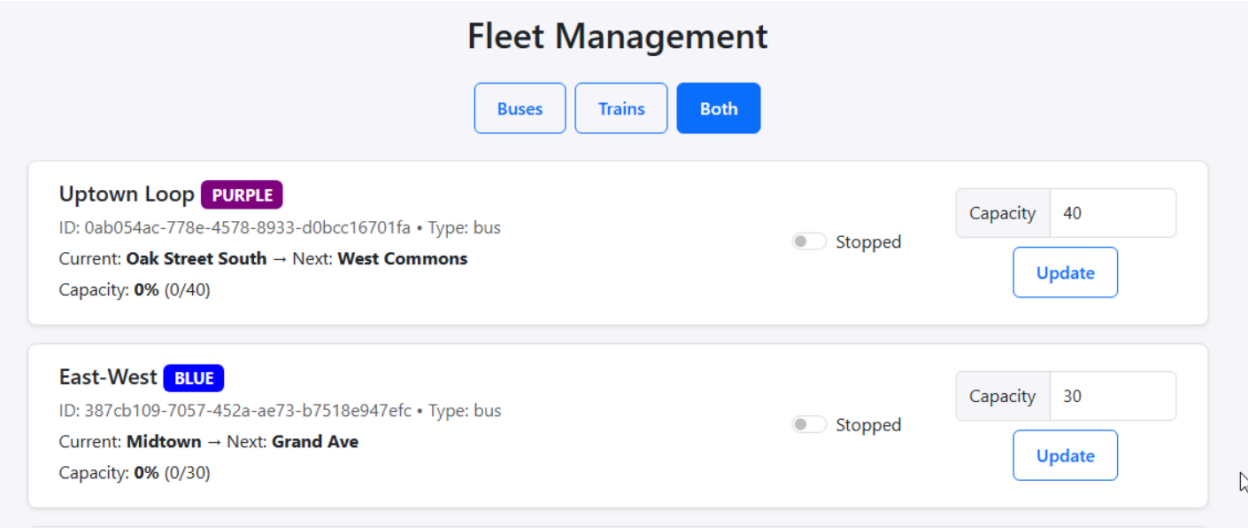


2. Similar to the website, a map of current bus positions is displayed on the home page. Below is also the status of each bus and train. Again, similar to the webpage this can be filtered by Bus, Train, or Both

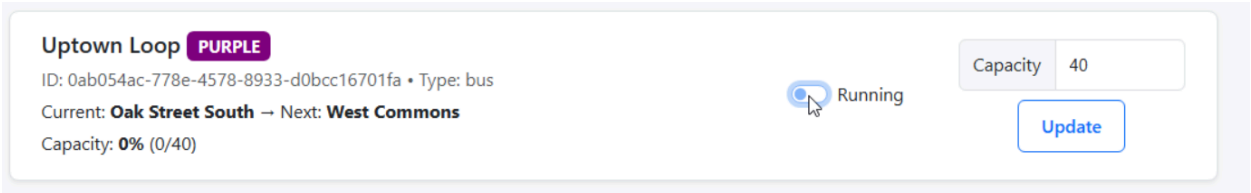


Updating vehicle status:

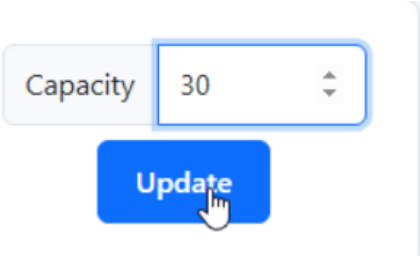
1. At the top navigation bar, select the 'Vehicles' option
2. You will see the vehicles in the fleet listed. This list can be filtered by busses only, trains only, or both



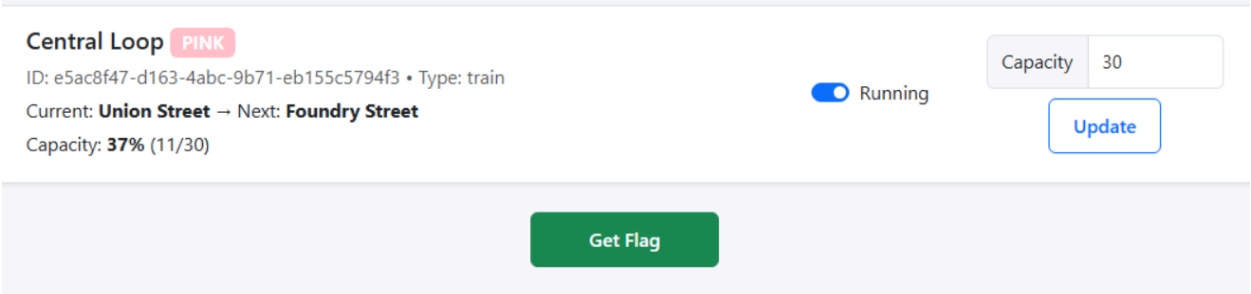
3. To enable/disable a vehicle, toggle the ‘stopped’ switch. Toggling this switch will automatically update the status of that vehicle



4. To add or reduce capacity, type a new value in the capacity box and click update to modify that vehicle’s capacity



5. At the bottom of the page is the ‘Get Flag’ button. When all buses or all trains are stopped, clicking this button will retrieve the required service flags. **Note:** This button only appears when the train only or bus only filter is selected at the top



Ticketing System

For testing purposes, an admin can make a curl request to the ticketing system from the Management Backend.

1. SSH into the Management Backend machine at the IP 89.65.43.30 following the steps on page 4.
2. Run the following command, replacing the values following subject, validFrom, and validUntil with the desired values:

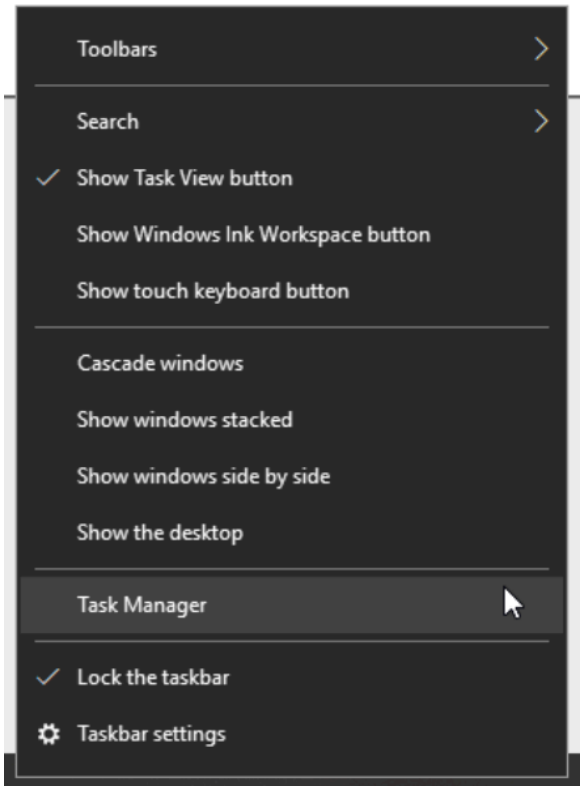
   curl -X 'POST' 'http://89.65.43.20:5000/api/tickets/validate' -H 'accept: */*' -H 'Content-Type: application/json' -d $(curl -X 'POST' 'http://89.65.43.20:5000/api/tickets/sign' -H 'accept: */*' -H 'Content-Type: application/json' -d '{"subject": "string", "validFrom": 1714000000, "validUntil": 2000000000}')

3. This will simultaneously call the sign and validate functions and will display the following response:

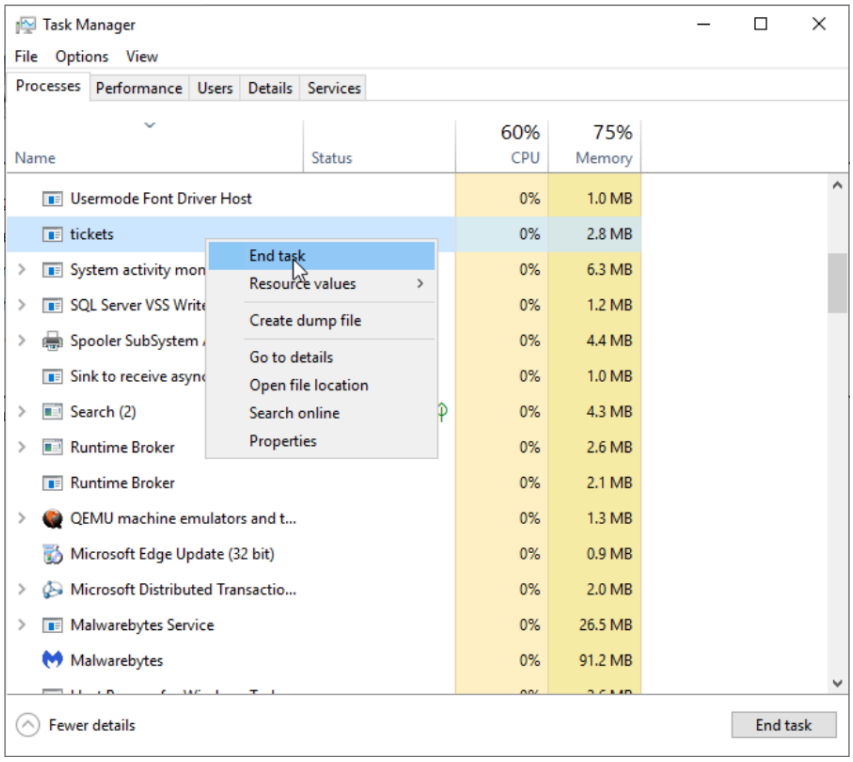
   {"valid":true,"reason":"OK","subject":"HokQHBFz094KCvCsVF9kzopazXWPbvAixQnTsrbvtbIHKmTFzQ","expiresAt":2000000000}

As an alternative, the ticketing system can be interacted with directly using the swagger dashboard:

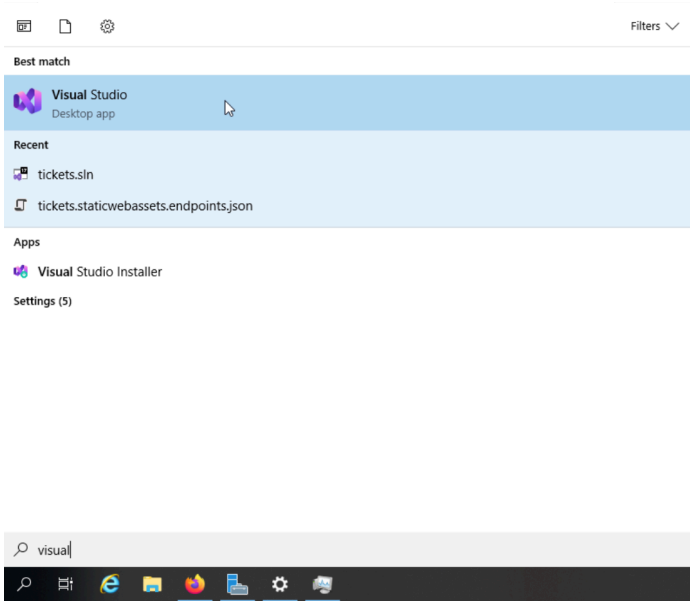
1. RDP into the Ticketing Machine at the IP 89.65.43.20 following the steps on page 2. You'll need to log in as an administrator.
2. Right click on the toolbar and select Task Manager



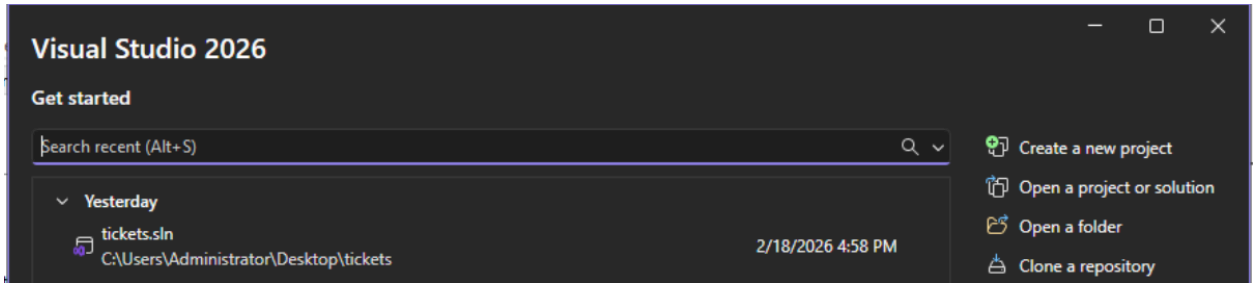
3. From Task Manager, locate the Process “tickets” and end it.



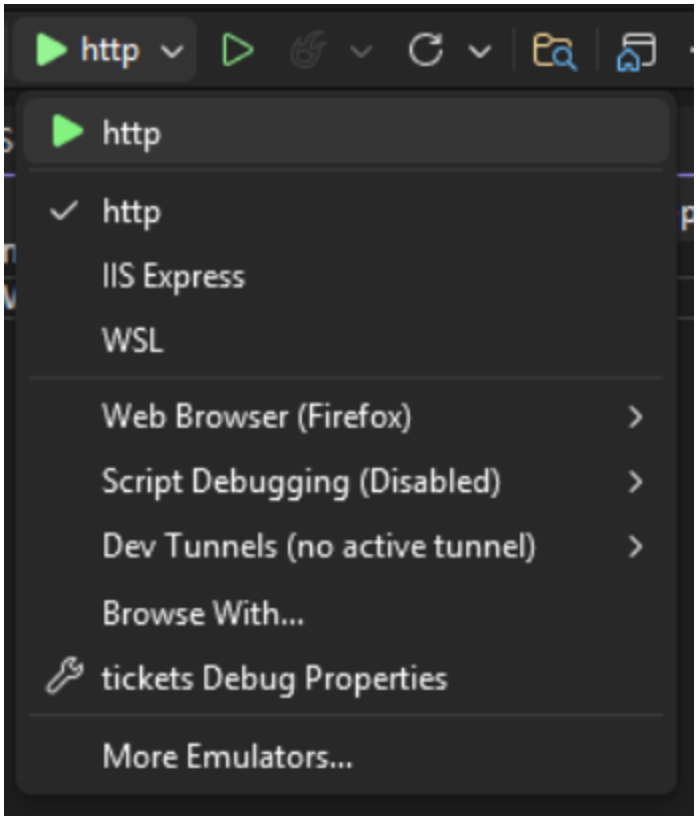
4. In Windows search, search for and open Visual Studio



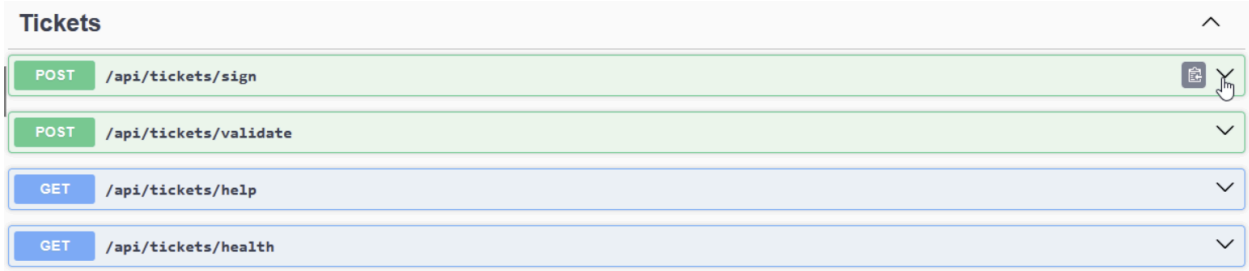
5. Locate the project called “tickets.sln” which has the file path C:\Users\Administrator\Desktop\tickets



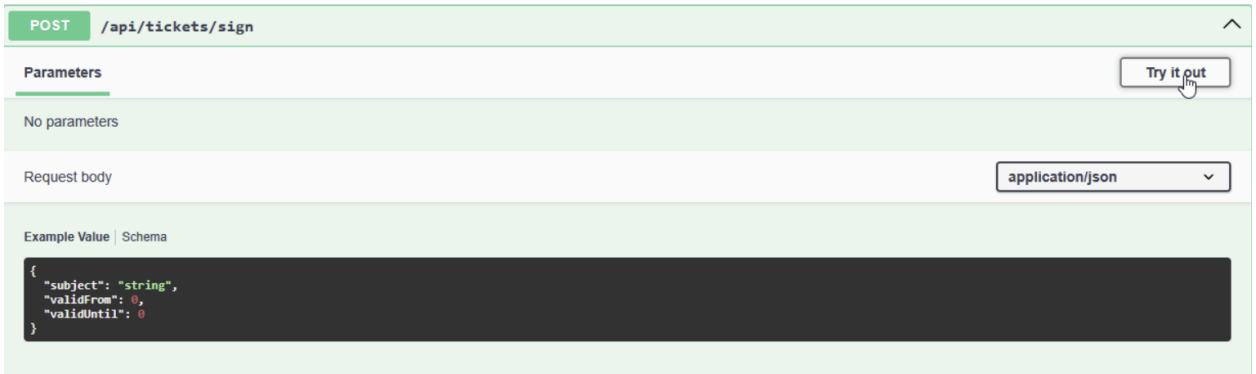
6. Within the project, verify that the correct project will run (should say http) and run it.



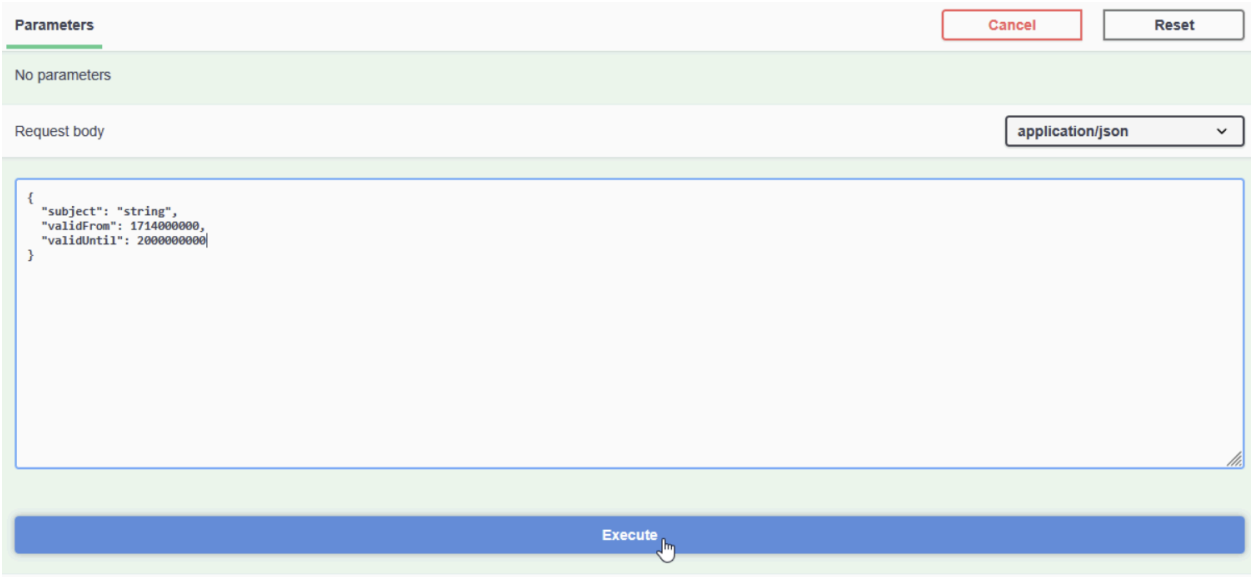
7. This should automatically open the desired site in FireFox, but if it does not proceed to http://89.65.43.20/swagger, which will pull up a page with the following options. Select the dropdown arrow on POST /api/tickets/sign.



8. Select the “Try it out” button.



9. Following the given format, fill in the “subject”, “validFrom”, and “validUntil” properties with the desired values and select the blue “Execute” button.



10. Copy the Response body which contains the information of the signed ticket.

Curl

```
curl -X 'POST' \
  'http://89.65.43.20:5000/api/tickets/sign' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "subject": "string",
  "validFrom": 1714000000,
  "validUntil": 2000000000
}'
```

Request URL

```
http://89.65.43.20:5000/api/tickets/sign
```

Server response

Code	Details
200	Response body


```
{
  "payload": "eyJleHAiOjIwMDAwMDAwMDAsImlhdCI6MTc3MTM3MTc5MywiaXNzIjoidGlja2V0LXNpZ251ciIsIm5iZiI6MTcxNDAwMDAwMCwic3ViIjoic3RyaW5nIn0=",
  "signature": "MwPD7W1f/aT04Vtp+MyHLNCULKqK8eu/VFIAvVPBt+HCgozy01CgDCwvjqZyj33WbrGyzeTJp0otGZXzhb4Tbjc6pqsUI0f2GwbOJhtrUV7LUSI69N1/kei5zg9qVBATSVLafM9CAJ0Uqfhoj2bCpE510TrZbu4GnOaqkRscs9XAv0dFEWN1pPZyGbmxCpZKkp0zL9HEDObPwT3GdgKTj3YE1w5F4yTmVSe3t+9iVAQlitbz+19fm1ktEnYo4xg7knNpG2rWFvMyR8r1khWa+4Ln0hb2AYHQzTysaMGjBF7UWu1iaYKwNebYp8DxM9udmAXxwIdZF/R92ih931uuVA==",
  "alg": "RS256",
  "kid": "5616999D608C57F8707825EE50A075D8EAF4174F"
}
```


Download

Response headers

```
content-type: application/json; charset=utf-8
date: Tue,17 Feb 2026 23:43:13 GMT
server: Kestrel
transfer-encoding: chunked
```

11. Follow the first two steps as before. Click the dropdown arrow on POST /api/tickets/validate and select “Try it out”.

POST /api/tickets/validate

GET /api/tickets/help

GET /api/tickets/health

POST /api/tickets/validate

Parameters

Try it out

No parameters

Request body

application/json

Example Value | Schema

```
{
  "payload": "string",
  "signature": "string",
  "alg": "string",
  "kid": "string"
}
```

12. The given format expects the exact response body from the signed ticket in step 4. Paste that information into the Request body and select the blue Execute button.

```
{
  "payload": "string",
  "signature": "string",
  "alg": "string",
  "kid": "string"
}
```

POST /api/tickets/validate

Parameters

Cancel | Reset

No parameters

Request body

application/json


```
{
  "payload": "eyJleHAiOjIwMDAwMDAwMDAsImlhdCI6MTc3MTM3MTc5MywiaXNzIjoidGlja2V0LXNpZ251ciIsIm5iZiI6MTcxNDAwMDAwMCwic3ViIjoic3RyaW5nIn0=",
  "signature": "MwPD7W1f/aT04Vtp+MyHLNCULKqK8eu/VFIAvVPBt+HCgozy01CgDCwvjqZyj33WbrGyzeTJp0otGZXzhb4Tbjc6pqsUI0f2GwbOJhtrUV7LUSI69N1/
kei5zg9qVBATSVLafM9CAJ0Uqfhoj2bCpE510TrZbu4GnOaqkRscs9XAv0dFEWW1pPZyGbmxCpZKkp0zL9HEDObPwT3GdgKTj3YE1w5F4yTmVSe3t+9iVAQlitbz+19fm1ktEnYo4xg7knNpG2rWFvMyR8rlkhWa+4Ln0hb2AYHQzTy
saMGjBF7UWu1iaYKwNebYp8DxM9udmAXxwIdZF/R92ih93luuVA==",
  "alg": "RS256",
  "kid": "5616999D608C57F8707825EE50A075D8EAF4174F"
}
```


Execute

Responses

Curl

```
curl -X 'POST' \
  'http://89.65.43.20:5000/api/tickets/validate' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "payload": "eyJleHAiOjIwMDAwMDAwMDAsImlhdCI6MTc3MTM3MTc5MywiaXNzIjoidGlja2V0LXNpZ251ciIsIm5iZiI6MTcxNDAwMDAwMCwic3ViIjoic3RyaW5nIn0=",
  "signature": "MwPD7W1f/aT04Vtp+MyHLNCULKqK8eu/VFIAvVPBt+HCgozy01CgDCwvjqZyj33WbrGyzeTJp0otGZXzhb4Tbjc6pqsUI0f2GwbOJhtrUV7LUSI69N1/kei5zg9qVBATSVLafM9CAJ0Uqfhoj2bCpE510TrZbu4GnO
  "alg": "RS256",
  "kid": "5616999D608C57F8707825EE50A075D8EAF4174F"
}'
```

Request URL

```
http://89.65.43.20:5000/api/tickets/validate
```

Server response

Code	Details
200	Response body

```
{
  "valid": true,
  "reason": "OK",
  "subject": "HokQHBFz094KCvCsVF9kzopazXWPbvAixQnTsrbvtbIHKmTFzQ",
  "expiresAt": 2000000000
}
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Tue,17 Feb 2026 23:46:38 GMT
server: Kestrel
transfer-encoding: chunked
```